

NETFLIX

1. Overview

This project involves a comprehensive analysis of Netflix's movies and TV shows data using SQL. The goal is to extract valuable insights and answer various business questions based on the dataset. The following README provides a detailed account of the project's objectives, business problems, solutions, findings, and conclusions.

2. Objectives

- Analyze the distribution of content types (movies vs TV shows).
- Identify the most common ratings for movies and TV shows.
- List and analyze content based on release years, countries, and durations.
- Explore and categorize content based on specific criteria and keywords.

3. Dataset

The data for this project is sourced from the Kaggle dataset:

4. Dataset Link:** [Movies Dataset](<https://www.kaggle.com/datasets/shivamb/netflix-shows?resource=download>)

5. Schema

```
```sql
DROP TABLE IF EXISTS netflix;
CREATE TABLE netflix
(
 show_id VARCHAR(5),
 type VARCHAR(10),
 title VARCHAR(250),
 director VARCHAR(550),
 casts VARCHAR(1050),
 country VARCHAR(550),
 date_added VARCHAR(55),
 release_year INT,
 rating VARCHAR(15),
 duration VARCHAR(15),
 listed_in VARCHAR(250),
 description VARCHAR(550)
);
```
```

6. Business Problems and Solutions

1. Count the Number of Movies vs TV Shows

```
```sql
SELECT
 type,
 COUNT(*)
FROM netflix
GROUP BY 1;
```
```

--Result:-

| | type  | count  |
|---|--|---|
| | text | bigint |
| 1 | Movie | 5185 |
| 2 | TV Show | 147 |

****Objective:**** Determine the distribution of content types on Netflix.

2. Find the Most Common Rating for Movies and TV Shows

```
```sql
WITH RatingCounts AS (
 SELECT
 type,
 rating,
 COUNT(*) AS rating_count
 FROM netflix
 GROUP BY type, rating
),
RankedRatings AS (
 SELECT
 type,
 rating,
 rating_count,
 RANK() OVER (PARTITION BY type ORDER BY rating_count DESC) AS rank
 FROM RatingCounts
)
SELECT
 type,
 rating AS most_frequent_rating
FROM RankedRatings
WHERE rank = 1;
```
```

--Result:-

| | type
text | most_frequent_rating
text |
|---|--------------|------------------------------|
| 1 | Movie | TV-MA |
| 2 | TV Show | TV-MA |

****Objective:**** Identify the most frequently occurring rating for each type of content.

3. List All Movies Released in a Specific Year (e.g., 2020)

```
```sql
SELECT *
FROM netflix
WHERE release_year = 2020;
```
```

--Result:--

| | show_id
text | type
text | title
text | release_year
bigint |
|---|-----------------|--------------|----------------------|------------------------|
| 1 | s2036 | Movie | The Social Dilemma | 2020 |
| 2 | s85 | Movie | Omo Ghetto: the Saga | 2020 |
| 3 | s127 | Movie | Shikara | 2020 |
| 4 | s228 | Movie | Really Love | 2020 |
| 5 | s230 | Movie | The Old Ways | 2020 |
| 6 | s283 | Movie | La diosa del asfalto | 2020 |
| 7 | s299 | Movie | Quam's Money | 2020 |
| 8 | s430 | Movie | Sanitation Day | 2020 |

****Objective:**** Retrieve all movies released in a specific year.

4. Find the Top 5 Countries with the Most Content on Netflix

```
```sql
SELECT *
FROM
(
 SELECT
 UNNEST(STRING_TO_ARRAY(country, ',')) AS country,
 COUNT(*) AS total_content
 FROM netflix
 GROUP BY 1
) AS t1
WHERE country IS NOT NULL
ORDER BY total_content DESC
LIMIT 5;
```
```

--Result:--

| | country
text | total_content
bigint |
|---|-----------------|-------------------------|
| 1 | United States | 2130 |
| 2 | India | 910 |
| 3 | United States | 355 |
| 4 | United Kingdom | 347 |
| 5 | Canada | 166 |

****Objective:**** Identify the top 5 countries with the highest number of content items.

5. Identify the Longest Movie

```
```sql
SELECT
 *
FROM netflix
WHERE type = 'Movie'
ORDER BY SPLIT_PART(duration, ' ', 1)::INT DESC;
```
```

--Result:-

| | show_id
text | type
text | title
text | duration
text |
|---|-----------------|--------------|-----------------------------|------------------|
| 1 | s2492 | Movie | The School of Mischief | 253 min |
| 2 | s2488 | Movie | No Longer kids | 237 min |
| 3 | s167 | Movie | Once Upon a Time in America | 229 min |
| 4 | s7933 | Movie | Sangam | 228 min |
| 5 | s1020 | Movie | Lagaan | 224 min |
| 6 | s4574 | Movie | Jodhaa Akbar | 214 min |
| 7 | s3228 | Movie | The Irishman | 209 min |
| 8 | s2732 | Movie | Kabhi Khushi Kabhie Gham | 209 min |

****Objective:**** Find the movie with the longest duration.

6. Find Content Added in the Last 5 Years

```
```sql
SELECT *
FROM netflix
WHERE TO_DATE(date_added, 'Month DD, YYYY') >= CURRENT_DATE - INTERVAL '5 years';
```
```

--Result:-

| | show_id
text | type
text | title
text | date_added
text |
|---|-----------------|--------------|-------------------------------|--------------------|
| 1 | s8 | Movie | Sankofa | September 24, 2021 |
| 2 | s9 | TV Show | The Great British Baking Show | September 24, 2021 |
| 3 | s10 | Movie | The Starling | September 24, 2021 |
| 4 | s13 | Movie | Je Suis Karl | September 23, 2021 |
| 5 | s25 | Movie | Jeans | September 21, 2021 |
| 6 | s28 | Movie | Grown Ups | September 20, 2021 |
| 7 | s29 | Movie | Dark Skies | September 19, 2021 |
| 8 | s30 | Movie | Paranoia | September 19, 2021 |

****Objective:**** Retrieve content added to Netflix in the last 5 years.

7. Find All Movies/TV Shows by Director 'Rajiv Chilaka'

```

```sql
SELECT *
FROM (
 SELECT
 *,
 UNNEST(STRING_TO_ARRAY(director, ',')) AS director_name
 FROM netflix
) AS t
WHERE director_name = 'Rajiv Chilaka';
```

```

--Result:-

| | show_id
text | type
text | title
text | director
text |
|---|-----------------|--------------|---------------------------------------|----------------------------|
| 1 | s411 | Movie | Chhota Bheem And The Broken Amulet | Rajiv Chilaka |
| 2 | s416 | Movie | Chhota Bheem aur Krishna | Rajiv Chilaka |
| 3 | s2718 | Movie | Chhota Bheem and the Curse of Damyaan | Rajiv Chilaka |
| 4 | s6465 | Movie | Chhota Bheem Kungfu Dhamaka | Rajiv Chilaka, Binayak Das |

****Objective:**** List all content directed by 'Rajiv Chilaka'.

8. List All TV Shows with More Than 5 Seasons

```

```sql
SELECT *
FROM netflix
WHERE type = 'TV Show'
 AND SPLIT_PART(duration, ' ', 1)::INT > 5;
```

```

| | title
text | duration
text |
|---|----------------------------------|------------------|
| 1 | The Great British Baking Show | 9 Seasons |
| 2 | The Flash | 7 Seasons |
| 3 | Men on a Mission | 6 Seasons |
| 4 | Call the Midwife | 9 Seasons |
| 5 | Supernatural | 15 Seasons |
| 6 | Arrow | 8 Seasons |
| 7 | Naruto | 9 Seasons |
| 8 | Comedians in Cars Getting Coffee | 6 Seasons |

****Objective:**** Identify TV shows with more than 5 seasons.

9. Count the Number of Content Items in Each Genre

```

```sql
SELECT
 UNNEST(STRING_TO_ARRAY(listed_in, ',')) AS genre,
 COUNT(*) AS total_content
FROM netflix
GROUP BY 1;
```

```

| | genre
text | total_content
bigint |
|---|---------------------------|-------------------------|
| 1 | International TV Shows | 45 |
| 2 | LGBTQ Movies | 1 |
| 3 | Stand-Up Comedy | 8 |
| 4 | Comedies | 426 |
| 5 | Spanish-Language TV Shows | 10 |
| 6 | TV Thrillers | 3 |
| 7 | Romantic Movies | 576 |
| 8 | Anime Features | 18 |

****Objective:**** Count the number of content items in each genre.

10. Find each year and the average numbers of content release in India on netflix.
return top 5 year with highest avg content release!

```
```sql
SELECT
 country,
 release_year,
 COUNT(show_id) AS total_release,
 ROUND(
 COUNT(show_id)::numeric /
 (SELECT COUNT(show_id) FROM netflix WHERE country = 'India')::numeric *
100, 2
) AS avg_release
FROM netflix
WHERE country = 'India'
GROUP BY country, release_year
ORDER BY avg_release DESC
LIMIT 5;
```
```

| | country
text | release_year
bigint | total_release
bigint | avg_release
numeric |
|---|-----------------|------------------------|-------------------------|------------------------|
| 1 | India | 2017 | 95 | 10.86 |
| 2 | India | 2018 | 79 | 9.03 |
| 3 | India | 2019 | 72 | 8.23 |
| 4 | India | 2016 | 69 | 7.89 |
| 5 | India | 2020 | 60 | 6.86 |

****Objective:**** Calculate and rank years by the average number of content releases by India.

11. List All Movies that are Documentaries

```
```sql
SELECT *
FROM netflix
WHERE listed_in LIKE '%Documentaries';
```

	show_id text	type text	title text	listed_in text
1	s2036	Movie	The Social Dilemma	Documentaries
2	s305	Movie	Lady Boss: The Jackie Collins Story	Documentaries
3	s717	Movie	FTA	Documentaries
4	s723	Movie	Sir! No Sir!	Documentaries
5	s758	Movie	Breaking Boundaries: The Science Of Our Planet	Documentaries
6	s791	Movie	Dream/Killer	Documentaries
7	s871	Movie	Amy Tan: Unintended Memoir	Documentaries
8	s915	Movie	Unrest	Documentaries

**\*\*Objective:\*\*** Retrieve all movies classified as documentaries.

## 12. Find All Content Without a Director

```
```sql
SELECT *
FROM netflix
WHERE director IS NULL;
```
```

|   | show_id<br>character varying (5) | type<br>character varying (10) | title<br>character varying (250)    | director<br>character varying (550) | casts<br>character varying |
|---|----------------------------------|--------------------------------|-------------------------------------|-------------------------------------|----------------------------|
| 1 | s2                               | TV Show                        | Blood & Water                       | [null]                              | Ama Qamata, Kh             |
| 2 | s4                               | TV Show                        | Jailbirds New Orleans               | [null]                              | [null]                     |
| 3 | s5                               | TV Show                        | Kota Factory                        | [null]                              | Mayur More, Jite           |
| 4 | s11                              | TV Show                        | Vendetta: Truth, Lies and The Mafia | [null]                              | [null]                     |
| 5 | s15                              | TV Show                        | Crime Stories: India Detectives     | [null]                              | [null]                     |
| 6 | s16                              | TV Show                        | Dear White People                   | [null]                              | Logan Browning,            |
| 7 | s369                             | TV Show                        | Khawatir                            | [null]                              | [null]                     |

**\*\*Objective:\*\*** List content that does not have a director.

## 13. Find How Many Movies Actor 'Salman Khan' Appeared in the Last 10 Years

```
```sql
SELECT *
FROM netflix
WHERE casts LIKE '%Salman Khan%'
AND release_year > EXTRACT(YEAR FROM CURRENT_DATE) - 10;
```
```

|   | type<br>character | title<br>character vary | casts<br>character varying (1050)                                                                                                                   |
|---|-------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Movie             | Paharganj               | Lorena Franco, Bijesh Jayarajan, Neet Chowdhary, Rajeev Gaursingh, Karran Jeet, Rajesh Sharma, Gautam Kurup, Poonam Mathur, Salman Khan, Manish ... |

**\*\*Objective:\*\*** Count the number of movies featuring 'Salman Khan' in the last 10 years.

## 14. Find the Top 10 Actors Who Have Appeared in the Highest Number of Movies Produced in India

```
```sql
SELECT
    UNNEST(STRING_TO_ARRAY(casts, ',')) AS actor,
    COUNT(*)
FROM netflix
WHERE country = 'India'
GROUP BY actor
ORDER BY COUNT(*) DESC
LIMIT 10;
```
```

|    | actor<br>text     | count<br>bigint |
|----|-------------------|-----------------|
| 1  | Anupam Kher       | 36              |
| 2  | Paresh Rawal      | 24              |
| 3  | Shah Rukh Khan    | 24              |
| 4  | Om Puri           | 23              |
| 5  | Akshay Kumar      | 22              |
| 6  | Boman Irani       | 21              |
| 7  | Amitabh Bachch... | 20              |
| 8  | Kareena Kapoor    | 19              |
| 9  | Naseeruddin Sh... | 19              |
| 10 | Asrani            | 17              |

**\*\*Objective:\*\*** Identify the top 10 actors with the most appearances in Indian-produced movies.

## 15. Categorize Content Based on the Presence of 'Kill' and 'Violence' Keywords

```
```sql
SELECT
    category,
    COUNT(*) AS content_count
FROM (
    SELECT
        CASE
            WHEN description ILIKE '%kill%' OR description ILIKE '%violence%'
        THEN 'Bad'
            ELSE 'Good'
        END AS category
    FROM netflix
) AS categorized_content
GROUP BY category;
```
```

|   | category<br>text | content_count<br>bigint |
|---|------------------|-------------------------|
| 1 | Good             | 8465                    |
| 2 | Bad              | 342                     |

**\*\*Objective:\*\*** Categorize content as 'Bad' if it contains 'kill' or 'violence' and 'Good' otherwise. Count the number of items in each category.

## 7. Findings and Conclusion

- **\*\*Content Distribution:\*\*** The dataset contains a diverse range of movies and TV shows with varying ratings and genres.
- **\*\*Common Ratings:\*\*** Insights into the most common ratings provide an understanding of the content's target audience.
- **\*\*Geographical Insights:\*\*** The top countries and the average content releases by India highlight regional content distribution.



- **\*\*Content Categorization:\*\*** Categorizing content based on specific keywords helps in understanding the nature of content available on Netflix.

This analysis provides a comprehensive view of Netflix's content and can help inform content strategy and decision-making.

**## Author - Ankul Analyst**

This project is part of my portfolio, showcasing the SQL skills essential for data analyst roles. If you have any questions, feedback, or would like to collaborate, feel free to get in touch!

- **\*\*LinkedIn\*\***: [Connect with me professionally](<https://www.linkedin.com/in/ankulkumarpatel56>)

Thank you for your support, and I look forward to connecting with you!